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CERTIFICATE

This is to certify that the project report titled “Footsteps Power Generation” is being submitted by **Jakkula Sharanya (24915A0410), Mudavath Sruthi (23911A0495), Gumpu Shiva Shanker(23911A0479), R Bheem reddy(23911A04A8)** of **III B. Tech II Semester, Department of Electronics and Communication Engineering**, as a record of bonafide work carried out by them. The results presented in this report are original and have not been submitted to any other University or institution for the award of any degree or diploma.

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DECLARATION

We hereby declare that the project work entitled “**Footsteps Power Generation**” is a genuine and original piece of work carried out by us in the **Department of Electronics and Communication Engineering, Vidya Jyothi Institute of Technology**, under the affiliation of **Jawaharlal Nehru Technological University, Hyderabad**. This report is the outcome of our own efforts and research conducted during the course of the project. We affirm that it has not been copied, reproduced, or submitted elsewhere for any academic or professional purpose. All sources of information and assistance have been duly acknowledged.

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ABSTRACT

The footstep power generation system operates on the principle of energy conversion, specifically transforming mechanical energy into electrical energy. When a person steps on piezoelectric sensors embedded in a surface, the applied mechanical stress produces an electrical potential due to the piezoelectric effect. The generated electrical output is typically low and fluctuating. Therefore, it is passed through a rectifier circuit to convert it from AC to DC, and then regulated to obtain a stable voltage. This conditioned energy is stored in a rechargeable battery or capacitor for later use.

The ESP32 microcontroller continuously monitors the generated and stored voltage through its analog input pins. It processes this data and displays the voltage values on an LCD screen in real time, allowing users to observe the energy generation. An LDR (Light Dependent Resistor) is integrated into the system to detect ambient light conditions. Based on the intensity of light, the ESP32 automatically controls a lighting system: In low light conditions, the light turns ON In bright conditions, the light turns OFF Additionally, the ESP32 enables IoT functionality by sending system updates, such as voltage levels and light status, to a remote platform like Telegram for monitoring. This system demonstrates an innovative approach to renewable energy generation by utilizing everyday human activity.

It is especially useful in high-traffic areas such as railway stations, shopping malls, and public walkways, where continuous foot movement can contribute to energy production. The integration of ESP32, LCD, and LDR enhances the system by adding real-time monitoring, automation, and remote communication capabilities

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CHAPTER-I

INTRODUCTION

Background and Importance

Background:

Footstep power generation is based on the concept of converting **mechanical energy into electrical energy** using piezoelectric sensors. Whenever pressure is applied (like walking), these sensors generate voltage. This idea comes under **energy harvesting systems**, which utilize unused energy from daily activities.

As mentioned in your abstract (page 3), this system uses **piezoelectric sensors and Arduino/ESP32** to capture and display generated energy in real-time.

Importance:

- Helps in **utilizing wasted human energy** effectively
- Provides a **renewable and eco-friendly energy source**
- Reduces dependence on conventional electricity
- Useful in **high footfall areas** like railway stations, malls, airports
- Supports **smart city and sustainable development concepts**
- Encourages innovation in **green energy technologies**.

Objectives and Purpose

Main Objective:

- To generate electrical energy from human footsteps using piezoelectric sensors.

Specific Objectives:

- To **capture mechanical energy** from footsteps using piezo plates
- To **convert pressure into electrical voltage**
- To **measure generated voltage** using a voltage sensor
- To **process data using ESP32 microcontroller**
- To **display output on LCD in real-time**
- To **store energy in a battery for later use**
- To implement **automatic street light control using LDR**
- To enable **remote monitoring via Telegram app**

Purpose:

- To demonstrate a **practical implementation of energy harvesting**
- To develop a **low-cost and efficient power generation system**
- To provide **power for small-scale applications** like LED street lights
- To integrate **IoT (ESP32 + Telegram)** for smart monitoring
- To promote **clean and sustainable energy solutions**

- To show how **everyday human activity can contribute to power generation**

Components Used and System Overview

The Footsteps Power Generation operates using three essential components:

- Piezoelectric sensors are used to convert mechanical pressure into electrical voltage.
- An ESP32 microcontroller acts as the central processing unit, handling data from sensors and controlling system operations.
- A voltage sensor measures the generated voltage, while an LCD display shows real-time output.
- An LDR (Light Dependent Resistor) is used to detect light conditions for automatic control of street lighting. A rechargeable battery stores the generated energy, and LEDs (street lights) act as the load.
- Additionally, the system uses Arduino IDE for programming and a Telegram app for remote monitoring and alerts.

The system works by converting human footsteps into electrical energy using piezoelectric sensors placed under a platform. When pressure is applied, voltage is generated and stored in a battery. This voltage is measured using a voltage sensor and sent to the ESP32 for processing. The processed data is displayed on an LCD screen in real time. Based on light conditions detected by the LDR sensor, the system automatically controls street lights by turning them ON or OFF. Additionally, the ESP32 enables IoT functionality by sending system data and alerts to the Telegram app for remote monitoring. This complete workflow is illustrated in the block diagram shown in your presentation

Significance and Applications

Significance:

Footstep power generation is significant because it introduces an innovative way of utilizing **wasted human energy** and converting it into useful electrical power. This system promotes **renewable and eco-friendly energy solutions**, reducing dependence on conventional power sources and supporting sustainable development. It is especially important in today's world where energy demand is increasing rapidly. The integration of sensors, microcontrollers, and IoT (ESP32 + Telegram) also highlights the role of **smart technologies in energy management**.

Applications:

The applications of footstep power generation are mainly found in **high footfall areas** where continuous human movement is available. It can be installed in places such as **railway stations, bus stops, airports, shopping malls, and colleges** to generate electricity from daily walking activity. The generated energy can be used for **street lighting, corridor lighting, display systems, and charging small devices** this technology can be integrated into **smart cities, public infrastructure, and IoT-based**

monitoring systems, making it a practical solution for sustainable energy generation in modern urban environments.

1.1 PROBLEM STATEMENT

The rapid growth in population and urbanization has led to a significant increase in the demand for electrical energy. Traditional energy sources such as coal, oil, and natural gas are being heavily consumed to meet this demand, which not only leads to resource depletion but also contributes to environmental pollution and global warming. This creates a major challenge in ensuring sustainable and clean energy for future generations.

At the same time, a large amount of energy produced through daily human activities, especially walking in crowded areas, is wasted. Every footstep exerts mechanical pressure on the ground, which contains potential energy that is not being utilized. In high footfall locations like railway stations, malls, and bus terminals, this unused energy represents a valuable opportunity for energy harvesting.

Existing energy generation methods are mostly centralized and depend on non-renewable resources, making them less efficient and environmentally harmful. There is a growing need for alternative solutions that are decentralized, renewable, and capable of generating power at the point of use. Harnessing small amounts of energy from multiple sources can collectively contribute to meaningful power generation.

To address this issue, piezoelectric technology can be used to convert mechanical stress from footsteps into electrical energy. However, designing a system that efficiently captures, stores, and utilizes this energy remains a challenge. It requires proper integration of sensors, energy storage devices, and control systems to ensure reliable operation.

Therefore, the problem is to develop an efficient, cost-effective, and eco-friendly system that can generate electrical energy from human footsteps using piezoelectric sensors. The system should be capable of storing the generated energy, monitoring its output, and utilizing it for practical applications such as lighting, while also incorporating smart features like automatic control and remote monitoring.

1.2 OBJECTIVES

- **Energy Generation from Footsteps**

The main objective of this project is to generate electrical energy from human footsteps using piezoelectric sensors and utilize it as a sustainable and eco-friendly power source for small-scale applications. The project aims to demonstrate how mechanical energy from daily human activity can be effectively converted into usable electrical energy.

- **Pressure Sensing and Energy Conversion**

Another objective is to capture the pressure produced by footsteps through piezoelectric plates and convert it into electrical voltage. This generated voltage is then measured using a voltage sensor to understand the efficiency of energy conversion. The system also focuses on ensuring that the generated energy can be stored in a rechargeable battery for future use.

- **Data Processing and Monitoring using ESP32**

The project further aims to process and monitor the generated data using the ESP32 microcontroller. The ESP32 acts as the brain of the system, enabling real-time data processing and control of various components. The measured voltage and system status are displayed on an LCD screen to provide a clear and continuous output to the user.

- **Automation using LDR Sensor**

An additional objective is to implement automation using an LDR sensor for street lighting. Based on the surrounding light conditions, the system automatically turns the lights ON during nighttime and OFF during daytime, improving energy efficiency and reducing manual intervention.

- **IoT Integration and Remote Monitoring**

Finally, the project aims to integrate IoT features by connecting the system to the Telegram app for remote monitoring and alerts. This allows users to track system performance and status from anywhere, making the solution smarter, more efficient, and suitable for modern smart city applications.

CHAPTER-2

LITERATURE SURVEY

Several research studies have been carried out in the field of **footstep power generation using piezoelectric sensors**, focusing on converting mechanical energy into electrical energy. Researchers have shown that piezoelectric materials generate voltage when subjected to pressure, making them suitable for energy harvesting from human footsteps. These systems are typically implemented in flooring structures where human movement is frequent, enabling continuous energy generation.

Many studies highlight the **basic working mechanism**, where piezoelectric sensors produce alternating voltage when pressure is applied. This voltage is then converted into usable DC power using rectifiers and capacitors, and stored in batteries for later use. Experimental setups often use multiple piezo sensors connected in series-parallel combinations to increase output efficiency and stabilize the generated voltage.

Research also emphasizes the **practical applications** of this technology in high footfall areas such as railway stations, shopping malls, schools, and airports. It has been observed that energy generated from continuous human movement can be used for low-power applications like lighting, charging devices, and sensors. This method is considered pollution-free and suitable for densely populated regions where human movement is abundant.

Recent advancements in this field include the integration of **IoT and smart monitoring systems**. Modern systems use microcontrollers like ESP32 along with communication technologies to monitor voltage levels and system performance remotely. Some research also explores optimization techniques and smart algorithms to improve efficiency and scalability for real-world urban applications.

Furthermore, studies on smart infrastructure suggest that piezoelectric tiles can be integrated into **public buildings and smart city designs** to create self-sustaining energy systems. Despite the promising results, challenges such as low power output, cost of implementation, and efficiency optimization still exist. Ongoing research is focused on improving materials, circuit design, and large-scale deployment of footstep energy harvesting systems.

2.1 LIMITATIONS OF EXISTING METHODOLOGY

The existing methodology, as shown in your presentation (page 5), uses piezoelectric sensors with Arduino to generate, process, and display energy from footsteps. However, this approach has several limitations. One of the major drawbacks is the **low power generation**, as the energy produced from a single footstep is very small and insufficient for large-scale applications.

Another limitation is the **inconsistent and fluctuating output voltage**, since the generated energy depends on the force and frequency of footsteps, which vary from person to person. This makes it difficult to obtain a stable power supply.

The system also suffers from **energy losses during conversion and storage**, especially in rectification and battery charging processes, reducing overall efficiency. Additionally, the existing setup mainly supports **small-scale applications only**, such as LEDs or basic displays, and cannot power heavy electrical loads.

Finally, the use of basic microcontrollers and simple circuitry limits advanced features like efficient power management, scalability, and real-time smart monitoring, making the system less effective for large-scale or commercial implementation.

CHAPTER-3

ANALYSIS

The analysis of the footstep power generation system shows that the amount of electrical energy generated depends directly on the **pressure applied and frequency of footsteps**. When higher force is applied on the piezoelectric plates, a higher voltage is produced, whereas lighter footsteps generate very low output. This indicates that the system performs better in areas with heavy and continuous foot traffic.

From the experimental observations, it is found that the generated voltage is **non-linear and fluctuating in nature**, making it necessary to use voltage sensors and processing units like ESP32 for monitoring and stabilization. The output is not constant, and therefore energy storage elements like batteries are essential to accumulate and utilize the generated power effectively.

The system is efficient for **low-power applications**, such as lighting LEDs or powering small electronic devices. However, due to limitations in energy generation and losses in conversion circuits, the overall efficiency remains moderate. Increasing the number of piezoelectric sensors and optimizing their arrangement can improve output performance.

The integration of ESP32 and IoT features enhances the system by enabling **real-time monitoring and control**, making it more suitable for smart applications. The use of LDR sensors further improves efficiency by automating street lighting based on environmental conditions.

Overall, the analysis concludes that while footstep power generation is a **promising renewable energy solution**, it is currently best suited for small-scale applications. With further improvements in materials, circuit design, and system optimization, it has the potential to become more efficient and widely applicable in smart city infrastructure.

3.1 FEATURES OF PROPOSED SYSTEM

The proposed footstep power generation system integrates piezoelectric sensors, ESP32 microcontroller, energy storage units, and IoT technology to achieve efficient energy harvesting and smart monitoring. This system is designed to generate renewable energy from human footsteps while ensuring real-time monitoring, automation, and cost-effectiveness. The key features of the proposed system are as follows:

- **Piezoelectric sensor-based energy generation**
The system employs piezoelectric sensors to convert mechanical pressure from human footsteps into electrical energy. These sensors generate voltage when force is applied, enabling effective energy harvesting from daily human activity.

- **Real-time voltage measurement and monitoring**
A voltage sensor is used to measure the generated electrical energy. The system continuously monitors voltage levels and provides accurate real-time data, ensuring proper analysis of energy generation efficiency.
- **ESP32-based processing and control**
The ESP32 microcontroller acts as the central unit, processing sensor data and controlling system operations. It ensures efficient data handling, system coordination, and integration of all components.
- **LCD display for live data visualization**
An LCD display is integrated to show real-time voltage values and system status. This provides a clear and user-friendly interface for monitoring system performance.
- **Automatic street light control using LDR**
The system uses an LDR sensor to detect ambient light conditions. Based on day and night detection, street lights are automatically turned ON and OFF, improving energy efficiency and reducing manual effort.
- **Energy storage and utilization**
The generated electrical energy is stored in a rechargeable battery. This stored energy can be used later for small-scale applications such as LED lighting and low-power devices.
- **IoT-based remote monitoring (Telegram integration)**
The system integrates IoT technology using ESP32 to send data and alerts to the Telegram app. This enables remote monitoring, making the system smart and accessible from anywhere.
- **Cost-effective and eco-friendly solution**
The system uses low-cost components and promotes renewable energy generation, making it environmentally friendly and suitable for educational and practical applications.
- **Scalability and future enhancement potential**
The proposed system can be further enhanced with advanced technologies such as improved energy storage systems, optimized sensor arrangements, and smart analytics for better efficiency and large-scale implementation.

This system demonstrates an innovative approach to utilizing human energy for power generation, supporting sustainable development and smart city applications.

3.2 ADVANTAGES OF PROPOSED SYSTEM

The footstep power generation system using piezoelectric sensors, ESP32, and IoT technology offers several advantages compared to conventional energy generation methods. Its design focuses on renewable energy utilization, cost-effectiveness, and smart automation, making it suitable for educational, research, and small-scale real-world applications. Some of the key advantages include:

- **Efficient utilization of wasted energy**
The system converts mechanical energy from human footsteps into electrical energy, ensuring that otherwise wasted energy is effectively utilized for useful applications.

- **Real-time monitoring capabilities**
The integration of ESP32 and voltage sensors enables real-time monitoring of generated energy. The data is displayed on an LCD and can also be accessed remotely, ensuring continuous system tracking.
- **Cost-effective implementation**
Unlike conventional power generation systems that require expensive infrastructure, this system uses low-cost components such as piezoelectric sensors and microcontrollers, making it affordable and accessible for students and researchers.
- **Automation with minimal manual intervention**
The system operates automatically by generating energy from footsteps and controlling street lights using an LDR sensor. This reduces the need for human involvement and improves efficiency.
- **Eco-friendly and pollution-free operation**
The system generates clean energy without producing harmful emissions, making it environmentally friendly and suitable for sustainable development.
- **Adaptability to different environments**
The system can be installed in various high footfall areas such as malls, railway stations, and colleges, making it adaptable to different environments where human movement is continuous.
- **Scalability for future enhancements**
The system provides a base model that can be improved with better energy storage, increased sensor efficiency, and integration with advanced technologies like IoT analytics and smart grids.
- **Practical learning and research application**
This project provides hands-on experience in energy harvesting, embedded systems, and IoT, making it highly useful for engineering students to understand real-world applications.

Overall, the proposed system combines simplicity, sustainability, and smart technology, making it an effective solution for small-scale renewable energy generation and smart city applications.

CHAPTER-4

SYSTEM DESIGN

The system design of the footstep power generation project is based on integrating energy harvesting, processing, storage, and automation into a single efficient unit. the system consists of piezoelectric sensors, voltage sensor, ESP32 microcontroller, battery, LCD display, LDR sensor, and lighting load. These components are interconnected to ensure smooth energy flow and real-time monitoring.

The design begins with the **energy generation unit**, where piezoelectric sensors are placed beneath a platform. When a person steps on the platform, mechanical pressure is applied to the sensors, generating an electrical voltage. This voltage is then passed through conditioning circuits (such as rectifiers) and stored in a rechargeable battery for later use.

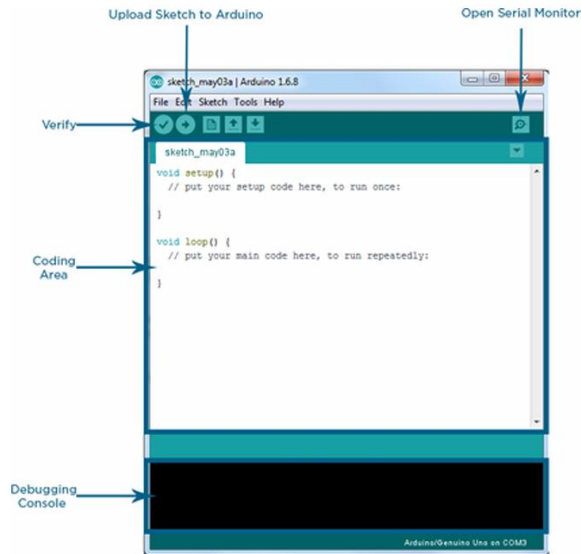
The **measurement and processing unit** includes a voltage sensor and ESP32 microcontroller. The voltage sensor measures the generated energy and sends the data to the ESP32. The microcontroller processes this data and manages the operation of other components. It acts as the central control system, ensuring proper coordination between sensing, display, and output.

The **display and control unit** consists of an LCD screen and an LDR sensor. The LCD displays real-time voltage and system status, while the LDR detects ambient light conditions. Based on this input, the ESP32 automatically controls street lighting, turning it ON during night and OFF during day.

Finally, the **communication and output unit** enables IoT functionality using ESP32. The system sends data and alerts to the Telegram app for remote monitoring. The stored energy in the battery is used to power LEDs or street lights, completing the overall system workflow. This design ensures an efficient, automated, and smart energy harvesting system suitable for real-world applications.

4.1 SOFTWARE REQUIREMENTS

The software requirements for the footstep power generation system include the necessary tools and platforms used for programming, monitoring, and communication. The primary software used is the **Arduino IDE**, which provides a simple environment to write, compile, and upload code to the ESP32 microcontroller. It supports programming using Embedded C/C++ and allows easy integration with various sensors and modules.



Software Name: Arduino IDE

Version: 1.8.x or later

Programming Language: C/C++ (Arduino language)

Purpose: Used to write, compile, and upload programs (sketches) to the Arduino board.

Key Features: Built-in serial monitor for data observation Predefined libraries for sensors and components Cross-platform (Windows, macOS, Linux) support User-friendly interface for debugging and code upload

To enable ESP32 functionality, the **ESP32 board support package (BSP)** must be installed in the Arduino IDE. Additional libraries such as **LiquidCrystal_I2C** (for LCD display), Wi-Fi libraries (for IoT communication), and sensor-related libraries are required for smooth operation of the system components. These libraries help in simplifying coding and hardware interfacing.

For IoT communication, the system uses the **Telegram application** along with a Telegram Bot API. This allows the ESP32 to send real-time data, alerts, and system updates directly to a mobile device. Proper configuration of Wi-Fi credentials and bot tokens is necessary to establish communication between the system and the user.

A **serial monitor** is used for debugging and testing during development. It displays real-time sensor readings, voltage values, and system logs, helping in identifying errors and ensuring correct functionality of the system.

Overall, the software setup is user-friendly, cost-effective, and well-suited for embedded system development, enabling efficient control, monitoring, and smart communication in the footstep power generation system.

4.2 HARDWARE REQUIREMENTS

The hardware requirements for the footstep power generation system include all the physical components needed to generate, process, store, and utilize electrical energy. The main component is the **piezoelectric sensor (plates)**, which converts mechanical pressure from footsteps into electrical voltage. Multiple piezo plates can be connected to increase the overall energy output.

The **ESP32 microcontroller** is used as the central processing unit, responsible for reading sensor data, controlling system operations, and enabling IoT communication. A **voltage sensor module** is used to measure the generated voltage from the piezo sensors and send it to the ESP32 for processing.

For displaying real-time data, a **16x2 LCD display with I2C module** is used, which shows voltage values and system status. An **LDR (Light Dependent Resistor)** is included to detect ambient light conditions and automatically control street lighting based on day and night.

A **rechargeable battery** is used to store the generated energy, ensuring that the power can be used later when needed. Additionally, **LED lights (street lights)** act as the load, utilizing the stored energy for illumination purposes.

Other supporting components include connecting wires, resistors, breadboard, and power supply modules, which are essential for proper circuit connections and system functioning. Overall, these hardware components work together to create an efficient and smart energy harvesting system.

4.3 FUNCTIONAL REQUIREMENTS

The functional requirements define the essential operations that the footstep power generation system must perform to ensure proper working and efficiency. The system should be capable of **detecting mechanical pressure from human footsteps** using piezoelectric sensors and converting it into electrical energy. It must continuously generate voltage whenever force is applied to the sensors.

The system should include a mechanism to **measure the generated voltage** using a voltage sensor and send this data to the ESP32 microcontroller for processing. The ESP32 should **analyze and manage the data**, ensuring proper coordination between all components. The processed data must be displayed in real-time on an LCD screen for user visibility.

The system must also be able to **store the generated energy in a rechargeable battery**, allowing the energy to be used later for practical applications such as lighting. It should ensure proper charging and safe storage of the generated power.

Another key requirement is **automatic control of street lighting using an LDR sensor**. The system should detect ambient light conditions and automatically turn the lights ON during nighttime and OFF during daytime without manual intervention.

Additionally, the system should support **IoT-based communication**, where the ESP32 sends real-time updates, alerts, and system status to the Telegram app. This enables remote monitoring and improves system usability. Overall, the system must operate efficiently, automatically, and reliably to achieve effective energy harvesting and smart control.

4.4 COMPONENTS USED IN PROPOSED SYSTEM

Arduino IDE

Arduino IDE is the primary software development environment used for programming the ESP32 microcontroller. It provides a user-friendly interface for writing, debugging, and uploading code, enabling smooth integration of sensors, harvesting units, and IoT features in the system.

Piezoelectric Sensor

Piezoelectric sensors are the core components responsible for energy generation. They convert mechanical pressure from human footsteps into electrical voltage. These sensors generate energy whenever force is applied, making them ideal for energy harvesting applications.

ESP32 Microcontroller

The ESP32 acts as the central control unit of the system. It processes the voltage data received from sensors, controls system operations, manages communication between components, and enables IoT-based monitoring through Wi-Fi connectivity.

LCD Display

The LCD display (16x2 with I2C module) is used to show real-time voltage values and system status. It provides a clear visual interface for users to monitor the energy generation process.

LDR (Light Dependent Resistor)

The LDR sensor detects ambient light conditions. It helps in automatically controlling street lights by turning them ON during nighttime and OFF during daytime, improving energy efficiency.

Battery

A rechargeable battery is used to store the generated electrical energy. This stored energy can be utilized later for powering small devices such as LEDs, ensuring that the generated power is not wasted.

LED / Street Light

LEDs act as the output load in the system. They utilize the stored energy to provide lighting, demonstrating the practical application of the generated power.

These components work together to create an efficient footstep power generation system, integrating energy harvesting, monitoring, storage, and smart automation for sustainable and real-time applications.

4.4.1 PIEZO ELECTRIC PLATE

A piezoelectric sensor is the most important component in the footstep power generation system, responsible for converting mechanical energy into electrical energy. It operates based on the **piezoelectric effect**, where certain materials such as quartz or ceramic generate an electric charge when subjected to mechanical stress. These sensors are placed beneath a platform so that whenever a person steps on it, pressure is applied, producing voltage.

Working Principle

The sensor consists of piezoelectric material sandwiched between conductive plates. When force is applied, the internal crystal structure deforms, causing charge separation and generating voltage across the electrodes. This voltage is usually alternating (AC) in nature and may require rectification before storage. The magnitude of generated voltage depends on the force applied and frequency of footsteps.

Role in System

Piezoelectric sensors act as the primary energy source in the system. They capture mechanical energy from human footsteps and convert it into electrical signals, which are further processed, measured, and stored. Multiple sensors are often connected in series or parallel to increase output efficiency.

Advantages of Piezoelectric Sensors

- Generates renewable energy from human activity
- Compact size and easy installation under flooring
- No need for external power supply
- Suitable for high footfall areas like malls and stations

Challenges and Limitations

- Generates very low power per step
- Output voltage is irregular and fluctuating
- Requires energy conditioning circuits for stability
- Efficiency depends on pressure and sensor placement

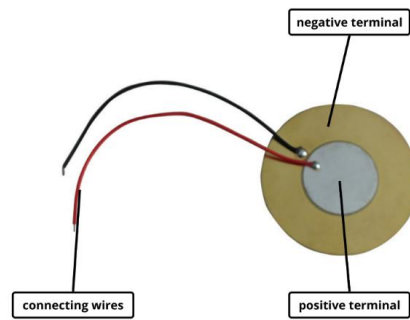


Fig. 4.4.1 Piezo Electric Plate

4.4.2 ESP32 MICROCONTROLLER

The ESP32 microcontroller is the central processing unit of the system, responsible for controlling all operations, processing data, and enabling communication. It is widely used due to its high performance, built-in Wi-Fi, and support for IoT applications.

Working Principle

The ESP32 reads analog voltage signals from sensors using its ADC (Analog-to-Digital Converter). It processes this data using programmed logic and controls output devices such as LCD displays and LEDs. It also connects to Wi-Fi networks for sending data to cloud platforms or mobile applications.

Role in System

ESP32 acts as the brain of the system. It collects voltage data, processes it, displays results, controls street lighting using LDR input, and sends updates to the Telegram app. It ensures coordination between all components.

Advantages

- High-speed processing and dual-core performance
- Built-in Wi-Fi and Bluetooth for IoT
- Low power consumption
- Supports multiple sensors and peripherals

Challenges and Limitations

- Requires proper coding and configuration
- Sensitive to voltage fluctuations
- Slightly complex compared to basic microcontrollers

Future Enhancements

- Use high-efficiency LEDs
- Integrate with smart lighting systems
- Adaptive brightness control using sensors



4.4.3 LED

4.4.4 BOARDBOARD

Description

A breadboard is a solderless prototyping platform used to build and test circuits easily. It allows components to be connected using jumper wires without permanent soldering.

Working Principle

Inside the breadboard, metal strips connect rows of holes. Terminal strips connect components, while bus strips distribute power. This allows easy circuit construction and modification.

Role in System

It is used for **prototyping and testing** the footstep power generation circuit before final implementation.

Advantages

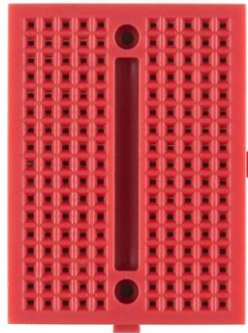
- No soldering required
- Reusable and flexible
- Easy debugging and modification

Challenges and Limitations

- Loose connections possible
- Not suitable for high current/frequency
- Limited durability

Future Enhancements

- Use PCB for permanent circuits
- Use simulation tools before prototyping
- Advanced modular prototyping boards



4.4.4 Breadboard

4.4.5 LDR SENSOR

Description

An LDR (Light Dependent Resistor) is a sensor that detects light intensity. Its resistance varies based on light falling on it.

Working Principle

In bright light, resistance decreases; in darkness, resistance increases. This change is converted into electrical signals for processing.

Role in System

It enables **automatic street lighting**, turning lights ON at night and OFF during daytime.

Advantages

- Simple and low cost
- Energy efficient automation
- Easy integration

Challenges and Limitations

- Affected by environmental light
- Slower response time
- Limited accuracy

Future Enhancements

- Replace with photodiodes for accuracy
- AI-based lighting control
- Smart adaptive lighting systems



4.4.5 LDR SENSOR

4.4.6 LCD I2C MODULE

Description

The LCD I2C module is used to display alphanumeric data using minimal wiring. It uses only two communication lines (SDA and SCL).

Working Principle

It operates using **I2C communication protocol**. The ESP32 sends data serially, and the I2C module converts it into parallel signals for the LCD.

Role in System

Displays **real-time voltage values, system status, and alerts** for monitoring.

Advantages

- Reduces wiring complexity
- Easy communication
- Low power consumption

Challenges and Limitations

- Limited display size
- Cannot show graphics
- Requires correct addressing

Future Enhancements

- Use OLED or TFT displays
- Graphical data visualization
- Touchscreen integration



4.4.6 LCD I2C MODULE

4.4.7 LI-ION BATTERY WITH TP4056 MODULE

A **Li-ion battery with TP4056 module** is used in the footstep power generation system for efficient energy storage and safe battery charging. The Li-ion battery stores the electrical energy generated by piezoelectric sensors, while the TP4056 module ensures proper charging, protection, and regulation of the battery. This combination is widely used in low-power energy harvesting systems due to its reliability and compact design.

Working Principle

The TP4056 is a linear battery charging IC designed specifically for **single-cell Li-ion batteries (3.7V)**. It operates using the **constant current–constant voltage (CC-CV)** charging method. Initially, it supplies a constant current to the battery, and once the battery reaches its threshold voltage (around 4.2V), it switches to constant voltage mode to safely complete charging. The module also includes protection circuits to prevent overcharging, over-discharging, and short circuits.

Role in System

In the footstep power generation system, the TP4056 module acts as an **interface between the generated energy and the battery**. The fluctuating voltage from piezo sensors is conditioned and fed into the TP4056, which safely charges the Li-ion battery. The stored energy is then used to power components like LEDs and ESP32 when required.

Advantages

- Safe charging with overcharge and discharge protection
- Compact and low-cost module

- Efficient energy storage for low-power systems
- Easy integration with energy harvesting circuits

Challenges and Limitations

- Supports only single-cell Li-ion batteries
- Requires stable input voltage for proper charging
- Not suitable for high-power applications
- Charging speed is limited for very low input energy

Future Enhancements

Future improvements may include integrating **boost converters** to stabilize low input voltage from piezo sensors, using **supercapacitors for faster energy storage**, and implementing **advanced battery management systems (BMS)** for improved efficiency and longer battery life.



Fig. 4.4.7 LI-ION BATTERY WITH TP4056 MODULE

CHAPTER-5

SYSTEM ARCHITECTURE

The system architecture for footstep power generation is designed to efficiently capture, process, store, and utilize energy generated from human footsteps. The system integrates piezoelectric sensors, a microcontroller (ESP32), energy storage units, display modules, and IoT communication. At the core of the system, piezoelectric sensors are placed beneath a platform to capture mechanical pressure when a person steps on it. These sensors convert mechanical energy into electrical energy in the form of voltage signals. Since the generated voltage is irregular and alternating, it is conditioned using rectification and filtering circuits before being used or stored.

The conditioned voltage is then measured using a voltage sensor and sent to the ESP32 microcontroller for processing. The ESP32 acts as the central control unit, analyzing the generated energy and managing system operations. It continuously monitors voltage levels, processes real-time data, and ensures proper coordination between input and output components. The processed data is displayed on an LCD screen, providing users with real-time information about energy generation and system status.

To enhance functionality, the system incorporates an LDR sensor for automation. Based on ambient light conditions, the ESP32 automatically controls street lighting by turning LEDs ON during night time and OFF during daytime. This improves energy efficiency and reduces unnecessary power consumption. Additionally, the system includes IoT capabilities, where the ESP32 sends data and alerts to the Telegram application, enabling remote monitoring and control.

Energy storage is a crucial part of the architecture. The generated energy is stored in a rechargeable Li-ion battery using a TP4056 charging module, ensuring safe and efficient charging. This stored energy can be used later to power LEDs or other low-power devices. Proper power management techniques are implemented to handle fluctuations in generated energy and maintain stable system operation.

By integrating energy harvesting, intelligent processing, automation, and communication, this system architecture provides a sustainable and smart solution for utilizing human-generated energy. It is suitable for applications in smart cities, public spaces, and energy-efficient infrastructure, demonstrating the potential of renewable micro-energy systems.

5.1 BLOCK DIAGRAM

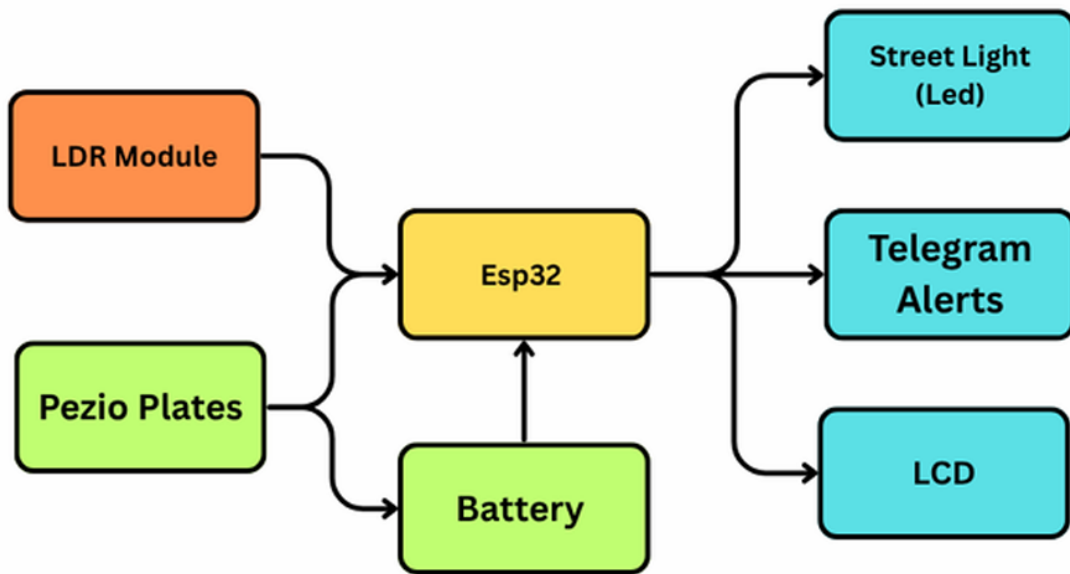


Fig. 5.1 Block diagram

This block diagram represents the **footstep power generation system using piezoelectric sensors** and illustrates the interaction between its core components.

- **LDR Module** – The LDR (Light Dependent Resistor) module is used to sense ambient light intensity. It continuously monitors environmental lighting conditions and sends corresponding signals to the ESP32. Based on this input, the system enables automatic control of street lighting, thereby improving energy efficiency.
- **Piezo Plates** – The piezoelectric plates function as the primary energy harvesting elements of the system. When mechanical pressure is applied through human footsteps, they generate electrical energy, which is then supplied to the system for further processing and storage.
- **Battery** – The battery serves as the energy storage unit of the system. It stores the electrical energy generated by the piezo plates and provides a stable and reliable power supply to the ESP32 and other components when required.
- **ESP32** – The ESP32 microcontroller acts as the central processing unit of the system. It receives inputs from the piezo plates and LDR module, processes the data, and controls all output operations. It ensures efficient coordination between energy generation, storage, and utilization while also enabling IoT-based communication.
- **Street Light (LED)** – The street light (LED) is the primary output device that utilizes the stored energy for illumination. It is automatically controlled by the ESP32 based on signals received from the LDR sensor, ensuring operation only during low-light conditions.
- **Telegram Alerts** – This component provides IoT-based communication by sending real-time notifications and system updates to the user through the Telegram application. It enables remote monitoring and enhances system accessibility.

- **LCD** – The LCD display is used to present real-time information such as generated voltage, battery status, and overall system performance. It provides a user-friendly interface for monitoring and analysis.

5.2 CIRCUIT DIAGRAM

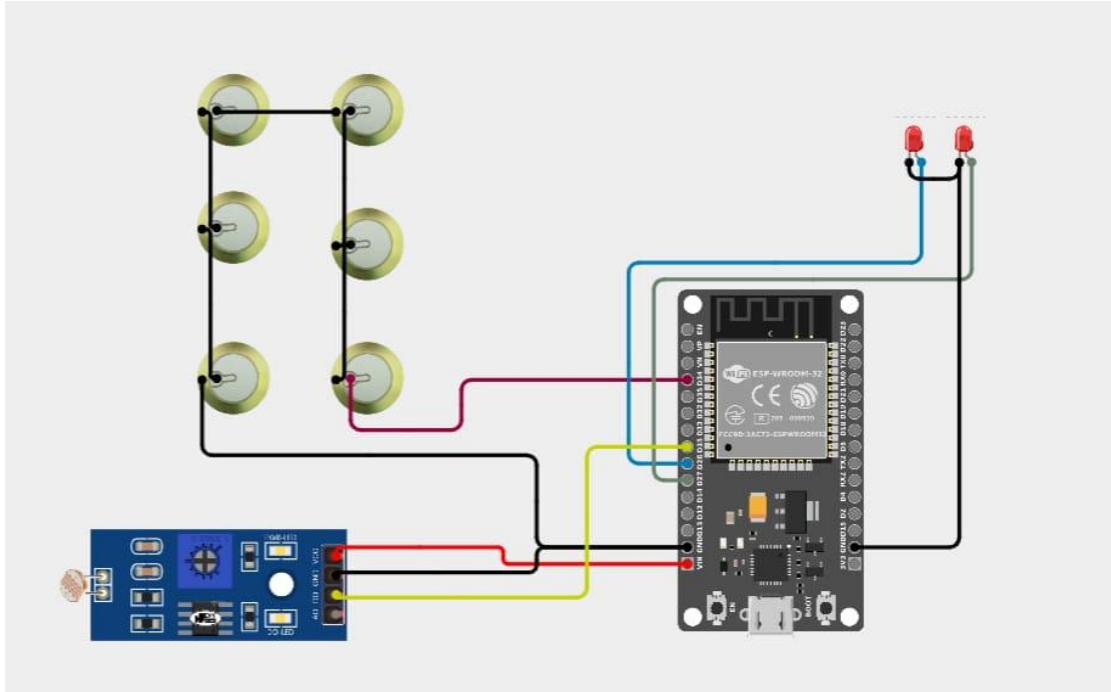


Fig. 5.2 Circuit diagram

The circuit diagram of the **footstep power generation system** represents the electrical connections between various components used to convert mechanical energy into electrical energy and utilize it effectively. The system mainly consists of piezoelectric sensors, rectifier circuit, voltage sensor, ESP32 microcontroller, battery with charging module, LCD display, LDR sensor, and LED load.

In this circuit, the piezoelectric sensors act as the input source, generating voltage when pressure is applied. The generated signal is conditioned and stored in a battery, while the ESP32 processes the data and controls the system operations. Additional components such as the LCD and LDR enhance monitoring and automation features.

Overall, the circuit diagram provides a clear understanding of how energy flows through the system and how different components interact to achieve efficient energy harvesting and smart control.

Piezoelectric Sensor Connections:

- **Positive Terminal** → Connected to rectifier circuit input to convert generated AC voltage into DC.
- **Negative Terminal** → Connected to system ground (GND).
- The output of the rectifier is connected to the voltage sensor and battery charging module.

ESP32 Connections:

- **VIN / 3.3V** → Connected to power supply or battery output.
- **GND** → Common ground connection for all components.
- **Analog Pin** → Receives voltage data from voltage sensor.
- **Digital Pins** → Connected to LCD, LDR, and LED for control and communication.

LCD (I2C Module) Connections:

- **VCC** → Connected to ESP32 5V/3.3V.
- **GND** → Connected to common ground.
- **SDA** → Connected to ESP32 SDA pin.
- **SCL** → Connected to ESP32 SCL pin.

LDR Sensor Connections:

- One terminal connected to **VCC**.
- Other terminal connected to **analog pin of ESP32** and resistor to GND (voltage divider).
- Provides light intensity input to ESP32.

Battery & TP4056 Module Connections:

- **IN+ / IN-** → Connected to rectified output from piezo sensors.
- **B+ / B-** → Connected to Li-ion battery terminals.
- **OUT+ / OUT-** → Supplies power to ESP32 and other components.

Piezoelectric Sensor Connections:

- **Positive Terminal** → Connected to rectifier circuit input to convert generated AC voltage into DC.
- **Negative Terminal** → Connected to system ground (GND).
- The output of the rectifier is connected to the voltage sensor and battery charging module.

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- **SCL** → Connected to ESP32 SCL pin.

LDR Sensor Connections:

- One terminal connected to **VCC**.
- Other terminal connected to **analog pin of ESP32** and resistor to GND (voltage divider).
- Provides light intensity input to ESP32.

Battery & TP4056 Module Connections:

- **IN+ / IN-** → Connected to rectified output from piezo sensors.
- **B+ / B-** → Connected to Li-ion battery terminals.
- **OUT+ / OUT-** → Supplies power to ESP32 and other components.

LED / Street Light Connections:

- **Anode (+)** → Connected to ESP32 digital output pin (via resistor).
- **Cathode (-)** → Connected to ground.

- **Anode (+)** → Connected to ESP32 digital output pin (via resistor).
- **Cathode (-)** → Connected to ground.

CHAPTER-6

METHODOLOGY

The methodology adopted for the project “**Footstep Power Generation using Piezoelectric Sensors**” follows a systematic approach to convert mechanical energy from human footsteps into electrical energy and utilize it efficiently. The approach focuses on integrating energy harvesting techniques with embedded systems and IoT-based monitoring. The entire development process can be broadly divided into three main phases: system design, hardware implementation, and testing.

Initially, a clear understanding of the **piezoelectric effect** was essential to select the core energy generation mechanism. Piezoelectric sensors generate electrical voltage when mechanical pressure is applied. Based on this principle, piezoelectric plates were chosen due to their simplicity, availability, and effectiveness in converting footstep pressure into electrical energy. The generated voltage is typically low and fluctuating, so appropriate signal conditioning methods such as rectification and filtering were considered necessary.

The next step involved interfacing the piezoelectric sensors with the processing unit, the **ESP32 microcontroller**. The ESP32 was selected due to its advanced features such as built-in Wi-Fi, sufficient input/output pins, and efficient processing capability. The voltage generated from the piezo sensors is passed through a voltage sensor module and then provided to the ESP32 for monitoring and processing.

Once the sensing and processing units were established, the system logic was developed around **real-time energy generation and monitoring**. Whenever a person steps on the piezo plates, voltage is generated and measured continuously. This data is processed by the ESP32 and displayed on an LCD screen, allowing users to observe the energy generation in real time. Simultaneously, the generated energy is directed to a **Li-ion battery through a TP4056 charging module**, ensuring safe storage of energy for later use.

To enhance system functionality, an **LDR sensor** was integrated for automation. The LDR continuously senses ambient light intensity and sends signals to the ESP32. Based on this input, the system automatically controls the LED street light, turning it ON during nighttime and OFF during daytime. This improves energy efficiency and eliminates the need for manual operation.

In addition to automation, the system incorporates **IoT-based communication** using the ESP32. The microcontroller is programmed to send real-time updates and alerts to a Telegram application. This enables remote monitoring of system performance, making the system more intelligent and user-friendly.

During hardware implementation, careful attention was given to circuit connections, voltage levels, and component integration. Proper wiring of piezo sensors, voltage sensor, ESP32, LCD, LDR, and battery module was ensured. A stable power supply and common grounding were maintained to avoid signal fluctuations and ensure reliable system operation.

After assembling the system, extensive testing was carried out to evaluate performance. Multiple trials were conducted by applying different levels of pressure on the piezo plates to observe voltage variations. Calibration of sensor readings and optimization of code parameters were performed to improve accuracy and efficiency. The automation feature using LDR and IoT communication was also tested under different environmental conditions.

The overall methodology focuses on developing a **cost-effective, sustainable, and smart energy harvesting system**. By combining renewable energy generation, embedded system control, automation, and IoT monitoring, the project demonstrates a practical approach to utilizing human-generated energy. Although the system is designed as a prototype, the methodology can be extended for large-scale applications in smart cities and public infrastructure.

In conclusion, this methodology highlights how fundamental principles of electronics, energy harvesting, and embedded systems can be integrated to create an efficient and intelligent system. It ensures effective utilization of wasted mechanical energy while promoting eco-friendly and sustainable solutions.

CHAPTER-7

IMPLEMENTATION

The implementation of the project “**Footstep Power Generation using Piezoelectric Sensors**” focuses on designing and developing a system that converts mechanical energy from human footsteps into electrical energy and utilizes it efficiently. The system integrates piezoelectric sensors, an ESP32 microcontroller, energy storage units, display modules, and IoT communication features. This chapter explains the step-by-step implementation process, including component integration, programming logic, circuit design, testing, and system performance.

Component Integration Strategy

The first step in implementation involved selecting appropriate components. Piezoelectric sensors were chosen as the primary energy harvesting elements due to their ability to generate voltage under mechanical stress. The ESP32 microcontroller was selected as the processing unit because of its high performance and built-in Wi-Fi capability. Additional components such as a voltage sensor, LCD display, LDR module, Li-ion battery, and TP4056 charging module were included to complete the system.

The piezoelectric sensors were connected in a suitable configuration (series/parallel) to enhance voltage output. The generated voltage was passed through a rectifier circuit to convert AC to DC and then connected to the TP4056 module for safe battery charging. The voltage sensor was interfaced with the ESP32 to measure the generated voltage. The LCD display was connected via I2C communication, while the LDR sensor was connected to an analog input pin of the ESP32. A common ground was maintained throughout the circuit to ensure stable operation.

Programming and Logic Flow :

The ESP32 was programmed using the Arduino IDE. The program begins with defining pin configurations and initializing components such as the LCD, Wi-Fi module, and sensors. In the setup function, communication protocols and input/output pins are configured.

In the loop function, the system continuously reads voltage values from the piezo sensors through the voltage sensor module. The ESP32 processes this data and displays it on the LCD in real time. Simultaneously, the LDR sensor reads ambient light intensity, and based on predefined threshold values, the ESP32 controls the LED street light accordingly.

Additionally, IoT functionality is implemented by connecting the ESP32 to a Wi-Fi network. The system sends real-time updates and alerts to the Telegram application, enabling remote monitoring. To improve accuracy, filtering techniques are applied in the code to reduce fluctuations in voltage readings.

Work Flow:

Step 1: Energy Generation Mechanical pressure is applied on piezoelectric plates, generating electrical energy.

Step 2: For Signal Conversion ,he generated AC voltage is converted into DC using a rectifier circuit.

Step 3: For Filtering and Storage,The DC voltage is filtered and stored in a rechargeable battery.

Step 4: For Voltage Monitoring The ESP32 reads the stored voltage using its analog input pin.

Step 5: Light Detection The LDR senses ambient light intensity. Low light indicates night, and high light indicates daytime.

Step 6: Street Light Control Based on LDR input, the ESP32 controls the LED. The light turns ON at night and OFF during the day.

Step 7: Data Processing The ESP32 processes voltage and sensor data continuously.

Step 8: Telegram Notification The ESP32 sends periodic updates including voltage level, LDR status, and LED status to a Telegram bot

Circuit Design and Prototyping:

Initially, the circuit was assembled on a breadboard for testing and validation. Each component was individually tested before integration. The piezoelectric sensors were tested for voltage generation under pressure. The voltage sensor readings were verified using the serial monitor. The LCD display was tested for proper output, and the LDR sensor was calibrated for light detection.

After successful testing, the components were arranged systematically to ensure compactness and proper connectivity. The circuit design included proper pin labeling, stable power supply connections, and protective components such as resistors and diodes where necessary. The TP4056 module was used to safely charge the battery, preventing overcharging and short circuits.

Testing and Validation:

The system was tested under various conditions to evaluate its performance. Different levels of pressure were applied to the piezoelectric sensors to observe variations in generated voltage. The accuracy of voltage measurement and display was verified. The response of the LDR sensor was tested under different lighting conditions to ensure proper automation of street lighting.

Multiple trials were conducted to check system stability and consistency. The IoT feature was tested by verifying message delivery to the Telegram app. Calibration was

performed by adjusting threshold values and delay timings to minimize fluctuations and improve system reliability.

The results showed that the system could successfully generate and store electrical energy from footsteps, display real-time data, and automate lighting based on environmental conditions.

Advantages and Limitations:

The implementation resulted in a cost-effective and eco-friendly system that demonstrates practical energy harvesting. The use of ESP32 allows integration of IoT features, making the system more advanced and user-friendly. The modular design enables easy expansion and future upgrades.

However, the system has certain limitations. The power generated from footsteps is relatively low and inconsistent, making it suitable only for low-power applications. Energy losses occur during conversion and storage. Additionally, the efficiency depends on the frequency and force of footsteps.

Despite these limitations, the system effectively demonstrates the concept of renewable energy generation and provides a foundation for further research and development.

7.1 SOURCE CODE

Arduino Code Implementation

```
#include <WiFi.h>

#include <WiFiClientSecure.h>

#include <UniversalTelegramBot.h>

#include <LiquidCrystal_I2C.h>


// WiFi credentials

const char* ssid = "HI";

const char* password = "000000000";


// Telegram credentials

#define BOT_TOKEN "8786612082:AAHKV_wKKqC-4QD2l2mu-ThMU_fKxWF5qYA"

#define CHAT_ID "6286745482"
```

```

WiFiClientSecure client;

UniversalTelegramBot bot(BOT_TOKEN, client);


// LCD setup
LiquidCrystal_I2C lcd(0x27, 16, 2);


// Pin definitions
#define VOLTAGE_PIN 34
#define LDR_PIN 25
#define LED_LDR 26
#define LED_VOLT 27


// Variables
float voltage = 0.0;
unsigned long lastTelegramTime = 0;
unsigned long previousMillisLED = 0;
const int ledBlinkInterval = 400; // Blink interval in ms


void setup() {
  Serial.begin(115200);

  pinMode(LDR_PIN, INPUT);
  pinMode(LED_LDR, OUTPUT);
  pinMode(LED_VOLT, OUTPUT);

  lcd.begin();
  lcd.backlight();

  // Improve ADC accuracy for low voltages
  analogSetAttenuation(ADC_11db);    // 0-3.3V full range

```

```

analogReadResolution(12);          // 12-bit ADC (0-4095)

WiFi.begin(ssid, password);
Serial.print("Connecting to WiFi");

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("Connected to WiFi");
client.setInsecure();

lcd.setCursor(0, 0);
lcd.print("System Ready");
delay(2000);
lcd.clear();
}

void loop() {

    // ----- Voltage Reading -----

    int total = 0;
    for (int i = 0; i < 50; i++) { // Average 50 samples for stability
        total += analogRead(VOLTAGE_PIN);
    }

    int sensorValue = total / 50;

    // Adjust factor based on your voltage sensor module
    voltage = (sensorValue / 4095.0) * 3.3 * 2;

```



```

Serial.print("Raw: ");
Serial.print(sensorValue);
Serial.print(" Voltage: ");
Serial.println(voltage);

// ----- LCD Display -----
lcd.setCursor(0, 0);
lcd.print("Voltage:");
lcd.setCursor(0, 1);
lcd.print(voltage, 2); // show 2 decimals
lcd.print(" V ");

// ----- Voltage LED Blink -----
unsigned long currentMillis = millis();
if (voltage > 0.05) { // detect low voltage threshold
    if (currentMillis - previousMillisLED >= ledBlinkInterval) {
        previousMillisLED = currentMillis;
        digitalWrite(LED_VOLT, !digitalRead(LED_VOLT)); // toggle
    }
} else {
    digitalWrite(LED_VOLT, LOW);
}

// ----- LDR Logic -----
int ldrState = digitalRead(LDR_PIN);
digitalWrite(LED_LDR, ldrState == HIGH ? HIGH : LOW);

// ----- Telegram Update -----
if (currentMillis - lastTelegramTime > 5000) { // every 5 sec

```

```

String message = "Status Update:\n";
message += "Voltage: " + String(voltage, 2) + " V\n";
message += "LDR: " + String(ldrState == LOW ? "Dark" : "Light") + "\n";
message += "LED: " + String(digitalRead(LED_LDR) ? "ON" : "OFF");

bot.sendMessage(CHAT_ID, message, "");
lastTelegramTime = currentMillis;
}

delay(200); // Small delay for loop stability
}

```

CHAPTER-8

RESULTS

The developed **footstep power generation system using piezoelectric sensors, ESP32, battery storage, and IoT integration** was tested under various conditions to evaluate its performance, efficiency, and reliability. This section presents the outcomes observed during testing and analyzes the effectiveness of each component in the system.

The primary objective of the project was to generate electrical energy from human footsteps and utilize it for small-scale applications such as LED lighting while monitoring the generated voltage in real time. During testing, the system successfully demonstrated the ability to convert mechanical energy into electrical energy whenever pressure was applied to the piezoelectric sensors. The generated voltage was displayed on the LCD, confirming real-time monitoring capability.

The piezoelectric sensors produced varying voltage levels depending on the force and frequency of footsteps. It was observed that higher pressure resulted in higher voltage output, while lighter steps produced relatively lower voltage. The voltage sensor accurately measured these variations and transmitted the data to the ESP32 for processing and display.

The energy storage system, consisting of a **Li-ion battery with TP4056 module**, functioned effectively by storing the generated energy. Although the amount of energy generated per step was small, continuous stepping resulted in gradual charging of the battery. The stored energy was successfully used to power LEDs, demonstrating practical utilization of harvested energy.

The automation feature using the **LDR sensor** was also tested under different lighting conditions. The system successfully turned the LED street light ON during low-light conditions and OFF during bright conditions, confirming proper functioning of automatic control. This feature enhanced energy efficiency by ensuring that power was used only when required.

The IoT functionality was evaluated by monitoring system updates through the Telegram application. The ESP32 successfully transmitted real-time data and alerts, enabling remote monitoring. This demonstrated the system's capability to integrate smart communication features with energy harvesting.

To evaluate performance, several test scenarios were conducted:

Test Case 1: Single Footstep Generation

When a single step was applied on the piezo plate, a small voltage spike was generated and displayed on the LCD. This confirmed basic functionality of energy generation.

Test Case 2: Continuous Footsteps

When multiple footsteps were applied continuously, the generated voltage increased

cumulatively, and the battery showed gradual charging. This demonstrated improved energy output under repeated usage.

Test Case 3: Light Intensity Variation

The LDR sensor was exposed to different lighting conditions. The system correctly switched the LED ON in darkness and OFF in light, confirming proper automation.

Test Case 4: IoT Communication Test

The system successfully sent real-time updates to the Telegram app, confirming proper wireless communication and remote monitoring capability.

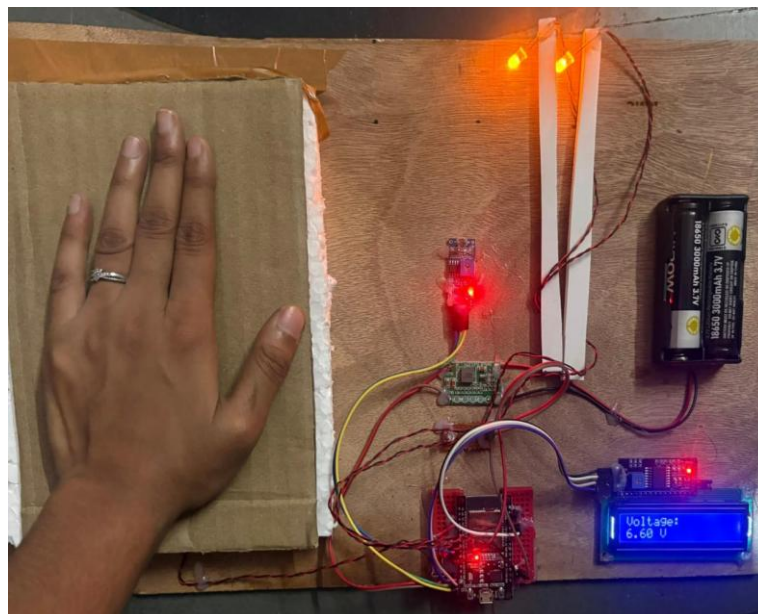
Test Case 5: Stability Test

The system was operated continuously for an extended period. It showed stable performance without major fluctuations or failures, indicating reliability.

In terms of hardware performance, the ESP32 efficiently processed sensor data and controlled all system operations without delay. The LCD displayed accurate and real-time information, and the battery charging module ensured safe energy storage. The piezoelectric sensors consistently generated voltage under applied pressure.

From a system perspective, the integration of energy harvesting, automation, and IoT communication was successful. The system responded accurately to input conditions and maintained stable operation throughout testing.

In conclusion, the results demonstrate that the footstep power generation system effectively achieves its objectives. It successfully converts mechanical energy into electrical energy, enables real-time monitoring, supports automation, and integrates smart communication features. Although the power output is limited, the system proves to be a viable and sustainable solution for small-scale energy applications and smart infrastructure development.



8.1: Hardware(Working Model)



8.2 : Telegram Bot

CHAPTER-9

CONCLUSIONS AND FUTURE SCOPE

Conclusion:

The project “**Footstep Power Generation using Piezoelectric Sensors**” successfully demonstrates the conversion of mechanical energy from human footsteps into electrical energy. The system effectively integrates piezoelectric sensors, ESP32 microcontroller, energy storage units, and IoT-based monitoring to create a functional and intelligent energy harvesting model.

The results confirm that the system is capable of generating electrical energy from everyday human activity and utilizing it for small-scale applications such as LED lighting. The implementation of real-time monitoring through LCD and remote communication via Telegram enhances the usability and smart functionality of the system. Additionally, the integration of an LDR sensor for automatic lighting control improves energy efficiency and reduces unnecessary power consumption.

Although the generated energy is relatively low and fluctuating, the project successfully highlights the potential of renewable micro-energy systems. It serves as a practical demonstration of sustainable energy concepts and shows how embedded systems and IoT technologies can be combined to develop innovative solutions. Overall, the project meets its objectives and provides a strong foundation for future advancements in energy harvesting systems.

Future Scope:

The future scope of the footstep power generation system lies in enhancing its efficiency, scalability, and real-world applicability. One of the major improvements can be achieved by using **advanced piezoelectric materials and nanogenerators**, which can significantly increase energy output.

Integration with **Artificial Intelligence (AI) and Machine Learning (ML)** can enable predictive analysis of foot traffic and optimize energy generation and usage. This can help in developing smarter and more adaptive systems for real-world applications.

The system can also be integrated with **smart grids and urban infrastructure**, allowing harvested energy to be distributed and utilized more effectively in public spaces such as railway stations, malls, and smart roads.

Further enhancements may include the use of **hybrid energy systems**, combining piezoelectric energy with solar power to increase overall efficiency. Advanced energy storage solutions such as supercapacitors and improved battery management systems can also be implemented.

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